Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 101506 | ***Subject: Mechanics of Materials*** | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) In infinitesimal strain theory, the square of displacement gradients is neglected because
(i) Materials remain rigid (iii) Deformations are very small
(ii) Stress remains constant (iv) Rotation is zero

(b) In thick cylinder analysis, Lame's equation assumes material to be
(i) Plastic and anisotropic (iii) Homogeneous and isotropic
(ii) Nonlinear elastic (iv) Viscoelastic

(c) A thick cylinder subjected to internal pressure experiences maximum hoop stress at
(i) Outer surface (iii) Mid-thickness
(ii) Inner surface (iv) Neutral axis

(d) Von-Mises criterion is primarily based on
(i) Maximum normal stress theory (iii) Shear strain energy theory
(ii) Maximum strain theory (iv) Coulomb friction theory

(e) The effective length factor for a column fixed at both ends is
(i) 2.0 (iii) 1.0
(ii) 0.7 (iv) 0.5

(f) The strain energy stored in a beam due to bending depends upon
(i) Shear modulus only (iii) Bending moment distribution
(ii) Axial load only (iv) Poisson's ratio only

(g) Plastic hinge formation in beams indicates
(i) Complete structural failure immediately (iii) Infinite stiffness
(ii) Yield moment has been reached (iv) Zero deflection condition

(h) Hill's yield criterion is mainly applicable to
(i) Brittle isotropic materials (iii) Ductile anisotropic materials
(ii) Elastic-perfectly plastic materials (iv) Composite beams only

(i) Shear center of a channel section lies
(i) At centroid (iii) Outside the section
(ii) At neutral axis only (iv) At geometric center

(j) Maxwell–Betti reciprocal theorem is valid under
(i) Plastic deformation (iii) Linear elastic behavior
(ii) Creep conditions (iv) Thermal instability

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Q.2 (a) State the assumption of theory of pure bending and derive the expression for bending stress in case of simple bending. [7]

(b) A simply supported beam of span 6 m carries Point load = 40 kN at center and Uniformly distributed load = 10 kN/m over entire span. Determine the central deflection and strain energy due to bending. Take EI=3×10^{12} N-mm^2. [7]

Q.3 (a) Write the assumption made in Euler's Formula for column and derive an expression of buckling load for column with one end fixed and the other end free. [7]

(b) A concrete pier 1.5 m × 1.5 m in section and 5 m long carries a compressive load 'P' lying at a point on the top which is 500 mm from each of the two adjacent faces. What is the maximum value of 'P', if no tensile stress is produced in the base? Take weight of concrete pier as 2.5×10^4 N/m^3. [7]

Q.4 (a) The state of stress at a point is given by: [7]

$$\sigma_x = x^2y,\ \tau_{xy} = xy,\ \sigma_y = y^2z,\ \tau_{yz} = yz,\ \sigma_z = z^2x,\ \tau_{xz} = xz$$

Determine the body force distribution at the point (1, 2, 3) so that the stresses are in equilibrium.

(b) The stress components at a point are: [7]

$$\sigma_x = 50,\ \tau_{xy} = 20,\ \sigma_y = 30,\ \tau_{yz} = 5,\ \sigma_z = 15,\ \tau_{xz} = 10 \text{ MPa}$$

with respect to xyz co-ordinate system. If the co-ordinate system is rotated about z-zxis in the anti-clockwise direction through 30°, Determine the new stress components.

Q.5 (a) Calculate the strain tensor for the following state of stress: [7]

$$\sigma_x = 100,\ \tau_{xy} = -500,\ \sigma_y = 50,\ \tau_{yz} = 50,\ \sigma_z = -200,\ \tau_{xz} = 0 \text{ MPa}$$

for a shear modulus of 80×10^6 kPa and an elastic modulus of 200×10^6 kPa.

(b) The components of strain tensor at a point in a body are: [7]

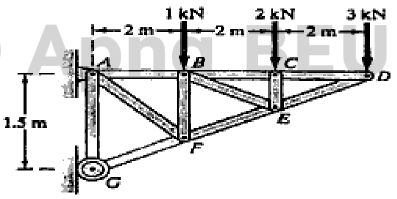
$\varepsilon_x = 0.01$, $\varepsilon_y = -0.005$, $\varepsilon_z = 0.005$, $\gamma_{xy} = 0.03$, $\gamma_{yz} = 0.01$, $\gamma_{xz} = -0.08$

Determine the principal strain.

Q.6 (a) Determine the diameter of a bolt which is subjected to an axial pull of 15 kN together with a transverse shear force of 7.5 kN using Tresca's theory. Take: elastic limit in tension = 225 N/mm^2, Factor of safety = 3 and Poisson's ratio = 0.3. [7]

(b) A cylindrical shell made of mild steel plate and 150 mm in diameter is to be subjected to an internal pressure of 15 N/mm^2. If the material yield at 200 MPa. Calculate the thickness of plate on the basis of maximum shear strain energy theory. Factor of safety = 3. [7]

Q.7 (a) Determine the force in member AF of the loaded truss as shown in figure. [7]



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(b) Explain the virtual work and unit load method for calculating deflection in beams and trusses. [7]

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Q.8 (a) Explain the stability conditions of retaining walls and chimneys against overturning and sliding with suitable equations. [7]

(b) Derive the differential relationships between load, shear force and bending moment and explain their application in beam analysis. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Castigliano's theorem
(b) Maxwell Bettie's reciprocal theorem
(c) Failure theories
(d) Stability of dams

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